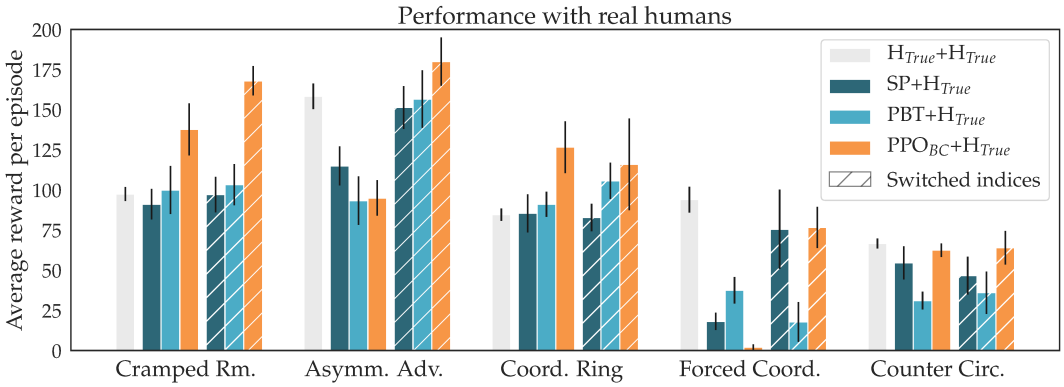

Figure 6: Average rewards over 400 timesteps of agents paired with real humans, with standard error across study participants. In most layouts, the PPO agent that trains with human data (PPO_{BC}, orange) performs better than agents that don't model the human (SP and PBT, teal), and in some layouts significantly so. We also include the performance of humans when playing with other humans (gray) for information. Note that for human-human performance, we took long trajectories and evaluated the reward obtained at 400 timesteps. In theory, humans could have performed better by optimizing for short-term reward near the end of the 400 timesteps, but we expect that this effect is small.

Due to computational complexity, model-based planning was only feasible on two layouts. Figure 5 shows the results on these layouts, demonstrating a similar trend. As expected, coupled planning achieves better self-play performance than reinforcement learning. But when pairing it with the human proxy, performance drastically drops, far below the gold standard. Qualitatively, we notice that a lot of this drop seems to happen because the agent expects optimal motion, whereas actual human play is much slower. Giving the planner access to the true human model and planning with respect to it is sufficient to improve performance (the dotted line above P_{BC}). However, when planning with BC but evaluating with H_{proxy}, the agent gets stuck in loops (see Appendix G).

Overall, these results showcase the benefit of getting access to a *good* human model – BC is in all likelihood closer to H_{Proxy} than to a real human. Next, we study to what extent the benefit is still there with a poorer model, i.e. when still using BC, but this time testing on real users.

6 User Study

Design. We varied the AI agent (SP vs. PBT vs. PPO_{BC}) and measured the average reward per episode when the agent was paired with a human user. We recruited 60 users (38 male, 19 female, 3 other, ages 20-59) on Amazon Mechanical Turk and used a between-subjects design, meaning each user was only paired with a single AI agent. Each user played all 5 task layouts, in the same order that was used when collecting human-human data for training. See Appendix H for more information.

Analysis. We present results in Figure 6. PPO_{BC} outperforms the self-play methods in three of the layouts, and is roughly on par with the best one of them in the other two. While the effect is not as strong as in simulation, it follows the same trend, where PPO_{BC} is overall preferable.

An ANOVA with agent type as a factor and layout and player index as covariates showed a significant main effect for agent type on reward ($F(2, 224) = 6.49$, $p < .01$), and the post-hoc analysis with Tukey HSD corrections confirmed that PPO_{BC} performed significantly better than SP ($p = .01$) and PBT ($p < .01$). This supports our hypothesis.

In some cases, PPO_{BC} also significantly outperforms human-human performance. Since imitation learning typically cannot exceed the performance of the demonstrator it is trained on, this suggests that in these cases PPO_{BC} would also outperform imitation learning.

We speculate that the differences across layouts are due to differences in the quality of BC and DRL algorithms across layouts. In Cramped Room, Coordination Ring, and the second setting of Asymmetric Advantages, we have both a good BC model as well as good DRL training, and so PPO_{BC} outperforms both self-play methods and human-human performance. In the first setting of Asymmetric Advantages, the DRL training does not work very well, and the resulting policy lets the human model do most of the hard work. (In fact, in the second setting of Asymmetric Advantages in

Figure 7: Cross-entropy loss incurred when using various models as a predictive model of the human in the human-AI data collected, with standard error over 5 different seeds. Unsurprisingly, SP and PBT are poor models of the human, while BC and H_{Proxy} are good models. See Appendix H for prediction accuracy.

Figure 4a, the human-AI team *beats* the AI-AI team, suggesting that the role played by the human is hard to learn using DRL.) In Forced Coordination and Counter Circuit, BC is a very poor human model, and so PPO$_{BC}$ still has an incorrect expectation, and doesn't perform as well.

We also guess that the effects are not as strong in simulation because humans are able to adapt to agent policies and figure out how to get the agent to perform well, a feat that our simple H_{Proxy} is unable to do. This primarily benefits self-play based methods, since they typically have opaque coordination policies, and doesn't help PPO$_{BC}$ as much, since there is less need to adapt to PPO$_{BC}$. We describe some particular scenarios in Section 7.

Figure 7 shows how well each model performs as a predictive model of the human, averaged across all human-AI data, and unsurprisingly finds that SP and PBT are poor models, while BC and H_{Proxy} are decent. Since SP and PBT expect the other player to be like themselves, they are effectively using a bad model of the human, explaining their poor performance with real humans. PPO$_{BC}$ instead expects the other player to be BC, a much better model, explaining its superior performance.

7 Qualitative Findings

Here, we *speculate* on some qualitative behaviors that we observed. We found similar behaviors between simulation and real users, and SP and PBT had similar types of failures, though the specific failures were different.

Adaptivity to the human. We observed that over the course of training the SP agents became very specialized, and so suffered greatly from distributional shift when paired with human models and real humans. For example, in Asymmetric Advantages, the SP agents only use the top pot, and ignore the bottom one. However, humans use both pots. The SP agent ends up waiting unproductively for the human to deliver a soup from the top pot, while the human has instead decided to fill up the bottom pot. In contrast, PPO$_{BC}$ learns to use both pots, depending on the context.

Leader/follower behavior. In Coordination Ring, SP and PBT agents tend to be very headstrong: for any specific portion of the task, they usually expect either clockwise or counterclockwise motion, but not both. Humans have no such preference, and so the SP and PBT agents often collide with them, and keep colliding until the human gives way. The PPO$_{BC}$ agent instead can take on both leader and follower roles. If it is carrying a plate to get a soup from the pot, it will insist on following the shorter path, even if a human is in the way. On the other hand, when picking which route to carry onions to the pots, it tends to adapt to the human's choice of route.

Adaptive humans. Real humans learn throughout the episode to anticipate and work with the agent's particular coordination protocols. For example, in Cramped Room, after picking up a soup, SP and PBT insist upon delivering the soup via right-down-interact instead of down-right-down-interact – even when a human is in the top right corner, blocking the way. Humans can figure this out and make sure that they are not in the way. Notably, PPO$_{BC}$ *cannot* learn and take advantage of human adaptivity, because the BC model is not adaptive.

8 Discussion

Summary. While agents trained via general DRL algorithms in collaborative environments are very good at coordinating with themselves, they are not able to handle human partners well, since they have never seen humans during training. We introduced a simple environment based on the game Overcooked that is particularly well-suited for studying coordination, and demonstrated quantitatively the poor performance of such agents when paired with a learned human model, and with actual humans. Agents that were explicitly designed to work well with a human model, even in a very naive way, achieved significantly better performance. Qualitatively, we observed that agents that learned about humans were significantly more *adaptive* and able to take on both *leader and follower roles* than agents that expected their partners to be optimal (or like them).

Limitations and future work. An alternative hypothesis for our results is that training against BC simply forces the trained agent to be robust to a wider variety of states, since BC is more stochastic than an agent trained via self-play, but it doesn't matter whether BC models real humans or not. We do not find this likely a priori, and we did try to check this: PBT is supposed to be more robust than self-play, but still has the same issue, and planning agents are automatically robust to states, but still showed the same broad trend. Nonetheless, it is possible that DRL applied to a sufficiently wide set of states could recoup most of the lost performance. One particular experiment we would like to run is to rain a single agent that works on arbitrary layouts. Since agents would be trained on a much wider variety of states, it could be that such agents require more general coordination protocols, and self-play-like methods will be more viable since they are forced to learn the same protocols that humans would use.

In contrast, in this work, we trained separate agents for each of the layouts in Figure 3. We limited the scope of each agent because of our choice to train the simplest human model, in order to showcase the importance of human data: if a naive model is already better, then more sophisticated ones will be too. Our findings open the door to exploring such models and algorithms:

Better human models: Using imitation learning for the human model is prone to distributional shift that reinforcement learning will exploit. One method to alleviate this would be to add inductive bias to the human model that makes it more likely to generalize out of distribution, for example by using theory of mind [7] or shared planning [17]. However, we could also use the standard data aggregation approach, where we periodically query humans for a new human-AI dataset with the current version of the agent, and retrain the human model to correct any errors caused by distribution shift.

Biasing population-based training towards humans: Agents trained via PBT should be able to coordinate well with any of the agents that were present in the population during training. So, we could train multiple human models using variants of imitation learning or theory of mind, and inject these human models as agents in the population. The human models need not even be accurate, as long as in aggregate they cover the range of possible human behavior. This becomes a variant of domain randomization [3] applied to interaction with humans.

Adapting to the human at test time: So far we have been assuming that we must deploy a static agent at test time, but we could have our agent adapt online. One approach would be to learn multiple human models (corresponding to different humans in the dataset). At test time, we can select the most likely human model [27], and choose actions using a model-based algorithm such as model predictive control [25]. We could also use a meta-learning algorithm such as MAML [11] to learn a policy that can quickly adapt to new humans at test time.

Humans who learn: We modeled the human policy as stationary, preserving the Markov assumption. However, in reality humans will be learning and adapting as they play the game, which we would ideally model. We could take this into account by using recurrent architectures, or by using a more explicit model of how humans learn.

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